

TNA as a Structural Framework for Transcendence, Sacramentality, and Non-Closure

Abstract

This paper explores the possibility that the Theory of Axiomatic Necessity (TNA), originally developed as a formal framework for realizability, boundary-induced collapse, and Failure of Local Closure in complex systems, may also provide a coherent structural interpretation of transcendence, sacramentality, and theological ontology.

The proposed framework distinguishes between two domains:

- N_0 : the domain of observable local dynamics, causal propagation, optimization, and material evolution;
- N_1 : the domain of external selection, legitimacy, realizability constraints, and non-local closure conditions.

Under TNA, sufficiently complex realizable systems may generate multiple admissible trajectories ($\Delta > 1$) without possessing a fully internal mechanism capable of selecting a unique stable realization. This condition — Failure of Local Closure — implies that complete ontological closure cannot always emerge from local dynamics alone.

The paper argues that several central Christian theological concepts — including the Logos, creatio continua, sacramentality, Eucharistic participation, and transcendence — can be reinterpreted structurally within this framework without reducing theology to physics or mathematics. Rather than eliminating mystery, mathematics here functions as a language revealing the limits of total derivability and closure.

Special attention is given to:

- the distinction between operational and structural closure,
- the continuous but non-smooth geometry of N_1 ,
- the Eucharist as participation in continuous realizability,
- and the difference between human openness and operationally closed AI systems.

The paper does not attempt to “prove” religious doctrine mathematically. Instead, it proposes that TNA may offer a structurally coherent metaphysical framework compatible with certain historical intuitions of classical Christianity regarding transcendence, Logos, and the non-self-sufficiency of being.

“The present work assumes familiarity with the TNA framework developed in previous publications concerning Failure of Local Closure, realizability constraints, and Boundary-Induced Collapse.”

1. Introduction

Modern technological civilization increasingly interprets reality through the lens of closure.

Within this paradigm:

- intelligence becomes optimization,
- consciousness becomes computation,
- meaning becomes emergent pattern recognition,
- and existence itself is interpreted as locally derivable from physical propagation.

Under sufficiently strong reductionism, transcendence appears unnecessary because all realizability is presumed derivable from internal material dynamics.

The Theory of Axiomatic Necessity (TNA) challenges this assumption.

TNA proposes that sufficiently complex realizable systems cannot always derive their own conditions of closure entirely from local internal dynamics. Under certain conditions, systems generate multiplicities of admissible trajectories without possessing a purely internal mechanism capable of selecting a unique stable realization.

This condition is called:

Failure of Local Closure.

The implications of this framework extend beyond economics, computation, or systems theory. If local closure is structurally incomplete, then transcendence may no longer appear as an arbitrary metaphysical addition, but rather as a structural requirement for realizability itself.

This paper explores the consequences of that possibility.

2. The Core Structure of TNA

TNA distinguishes between two structural domains:

2.1 N_0 : Local Dynamics

N_0 denotes:

- observable physical processes,
- optimization,
- causal propagation,
- biological evolution,
- computation,
- and operational dynamics.

N_0 is characterized by:

- local continuity,
- differentiability,
- propagation rules,
- and internal state evolution.

Physics, economics, machine learning, and biological adaptation all operate primarily within N_0 .

2.2 N_1 : External Selection and Realizability

N_1 denotes:

- external selection,
- legitimacy,
- realizability constraints,
- boundary conditions,
- and non-local closure operators.

N_1 does not replace N_0 .

Rather, it constrains realizability itself.

Within TNA, N_1 becomes necessary whenever:

- $\Delta > 1$,
and
 - local dynamics alone cannot uniquely determine realization.
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2.3 Failure of Local Closure

A system experiences Failure of Local Closure when:

- multiple admissible trajectories exist,
- but no purely local mechanism can uniquely select among them.

This condition appears across:

- economics,
 - social systems,
 - institutional collapse,
 - morality,
 - consciousness,
 - and potentially metaphysics itself.
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3. The Logos as Structural Intelligibility

Gospel of John

The opening of the Gospel of John introduces the Logos not merely as speech, but as the principle through which existence itself becomes coherent:

■ *“In the beginning was the Word.”*

Historically, Logos implied:

- intelligibility,
- order,
- rationality,

- structure,
- and meaningful coherence.

Under TNA, the Logos may be interpreted structurally as:

- the intelligible support of realizability.

This interpretation does not reduce theology to mathematics.
Rather, it suggests that:

- intelligibility itself may be inseparable from the conditions of realizability.
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4. Continuous Creation and Non-Closure

Classical Christian theology historically maintained the doctrine of *creatio continua*: creation is continuously sustained rather than merely initiated in the distant past.

TNA provides a structural interpretation of this intuition.

If realizability cannot fully derive itself from local closure alone, then reality cannot be interpreted as a perfectly autonomous machine.

The universe remains:

- dynamically continuous,
yet
 - structurally open.
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5. The Geometry of N_1

The present framework proposes that:

- N_0 behaves locally as a smooth differentiable domain,
while
- N_1 remains continuous but globally non-smooth.

Formally:

- continuous (C^0),

- but not fully differentiable (non- C^1).

This distinction is critical.

If N_1 were fully differentiable, then local dynamics could eventually extrapolate and derive all future realizations internally.

In that case:

- transcendence collapses into determinism,
- novelty disappears,
- and complete closure becomes possible.

Non-smoothness preserves:

- openness,
 - novelty,
 - freedom,
 - and realizability bifurcation.
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6. Suffering, Discontinuity, and Structural Rupture

Within this framework:

- suffering,
- existential rupture,
- historical collapse,
- and moral bifurcation

can be interpreted as manifestations of structural non-closure.

This does not morally justify suffering.

Rather, TNA proposes that:

- discontinuity may be structurally unavoidable in realizable open systems.

A perfectly smooth universe would imply complete derivability.

A completely discontinuous universe would eliminate coherence.

Reality instead appears:

- locally smooth,
 - globally open,
 - and structurally incomplete.
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7. Sacramentality and Eucharistic Participation

Council of Trent

Catholic sacramental theology historically maintained that the Eucharist is not merely symbolic remembrance, but real participation.

Under TNA, the Eucharist may be interpreted structurally as:

- participation in continuous realizability.

The human being does not remain ontologically self-grounded.

Instead:

- existence remains continuously sustained by relation to the Logos.

The Eucharist becomes:

- ritual acknowledgment of non-closure.

To “receive the Logos” is not merely psychological symbolism.

It is acceptance that:

- the self cannot fully derive itself from local material propagation alone.

“The Eucharistic transformation preserves observable local continuity (N_o) while altering the ontological support of realizability through relation to N_i .”

“The Incarnation represents not merely divine intervention within history, but the explicit realizability intersection between N_o and N_i .”

8. Human Openness and Artificial Intelligence

Current AI systems operate within operationally closed state spaces.

Even highly advanced systems:

- optimize,
- self-modify,
- and generate multiplicity internally,
while remaining constrained by predefined architectures and admissible operational spaces.

Human beings appear structurally different.

Humans:

- sacrifice utility,
- pursue transcendence,
- reject optimization,
- and question the legitimacy of the state space itself.

This distinction may reflect:

- operational closure in AI,
versus
- structural openness in human consciousness.

Under TNA:

AI may possess operational multiplicity without possessing structural openness toward N_1 .

9. Mathematics and the Limits of Closure

The problem is not the use of mathematics.

The real problem appears when mathematics becomes:

- an ideology of total closure.

If mathematical formalization is interpreted as implying complete derivability, then transcendence, freedom, and novelty disappear.

Paradoxically, TNA uses mathematics to argue precisely the opposite:

- reality may be structurally incapable of complete self-closure.

In this sense, mathematics becomes:

- a language of intelligibility,
not
 - a mechanism for eliminating mystery.
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10. Transcendence as Structural Requirement

Within strongly reductionist paradigms, transcendence appears unnecessary because local dynamics are presumed sufficient.

TNA challenges that assumption.

If realizability itself requires conditions not fully derivable from local propagation, then transcendence ceases to appear merely mythological.

Instead, transcendence becomes:

- structurally compatible with the ontology of realizable open systems.

This does not prove theology mathematically.

But it suggests that:

- certain theological intuitions may correspond to genuine structural features of realizability.
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11. Conclusion

This paper proposed that the Theory of Axiomatic Necessity may provide a coherent structural framework for interpreting:

- transcendence,
- sacramentality,
- Logos ontology,

- and continuous creation.

The framework does not attempt to reduce theology to mathematics or physics. Rather, it argues that:

- mathematics itself may reveal the limits of total closure.

Under TNA:

- local dynamics remain real,
- science remains valid,
- causality remains meaningful,
yet
- complete ontological closure may remain impossible.

Reality therefore appears:

- locally derivable,
- globally open,
- and continuously sustained.

The Eucharist, under this interpretation, becomes more than symbolic ritual: it becomes participation in continuous realizability itself.

The human being emerges not as a closed optimization machine, but as an open system structurally oriented toward transcendence.

In this sense, TNA suggests that mystery is not eliminated by intelligibility.

Rather:

- intelligibility itself may require openness beyond purely local closure.

"Mathematical intelligibility does not eliminate the mystery; it reveals structurally why complete closure might be impossible."

References

Thomas Aquinas

Aquinas, T. Summa Theologica.

Augustine of Hippo

Augustine. Confessions.

Kurt Gödel

Gödel, K. *On Formally Undecidable Propositions of Principia Mathematica and Related Systems.*

Pierre Teilhard de Chardin

Teilhard de Chardin, P. *The Phenomenon of Man.*

Alfred North Whitehead

Whitehead, A. N. *Process and Reality.*

Gospel of John

Irenaeus

Athanasius of Alexandria

Maximus the Confessor

Bresciano, C. *Failure of Local Closure and Boundary-Induced Collapse.* Zenodo. DOI: 10.5281/zenodo.18644522.

Bresciano, Claudio. *The Theorem Nobody Wants: Metamathematical Necessity of Axioms via Reduction to Incoherence v1.3.* Zenodo. DOI 10.5281/zenodo.19192655

Bresciano, Claudio. *Randomness as Perceived Non-Uniqueness: A Structural Interpretation of Indeterminacy in Non-Lipschitz Systems.* Zenodo. DOI 10.5281/zenodo.19925656